Class 18

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background

pertussis is a common lung infection caused by the bacteria b.pertussis.

this can affect people of any age, but affect the most for infants

the cdc track the number of reported ases in US (https://www.cdc.gov/pertussis/php/surveillance/pertussis-cases-by-year.html?CDC_AAref_Val=https://www.cdc.gov/pertussis/surv-reporting/cases-by-year.html):

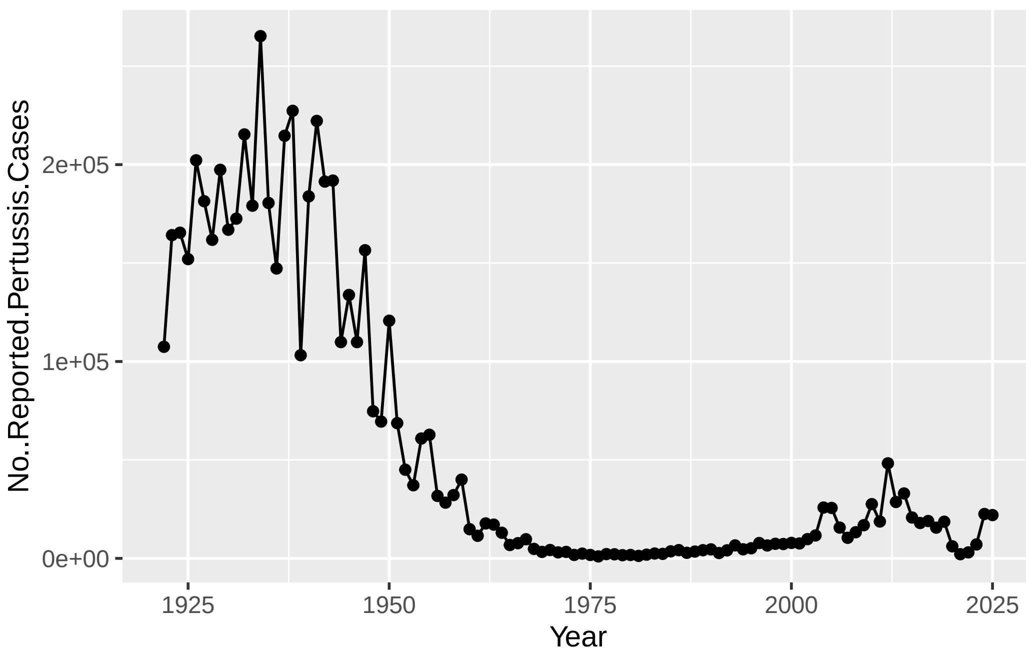
We can "scarpe" this data with the **datapasta** package

```
cdc <-  data.frame(
                                  Year = c(1922L,1923L,1924L,1925L,
                                           1926L,1927L,1928L,1929L,1930L,
                                           1931L,1932L,1933L,1934L,1935L,
                                           1936L,1937L,1938L,1939L,1940L,1941L,
                                           1942L,1943L,1944L,1945L,1946L,
                                           1947L,1948L,1949L,1950L,1951L,
                                           1952L,1953L,1954L,1955L,1956L,1957L,
                                           1958L,1959L,1960L,1961L,1962L,
                                           1963L,1964L,1965L,1966L,1967L,
                                           1968L,1969L,1970L,1971L,1972L,
                                           1973L,1974L,1975L,1976L,1977L,1978L,
                                           1979L,1980L,1981L,1982L,1983L,
                                           1984L,1985L,1986L,1987L,1988L,
                                           1989L,1990L,1991L,1992L,1993L,1994L,
                                           1995L,1996L,1997L,1998L,1999L,
                                           2000L,2001L,2002L,2003L,2004L,
                                           2005L,2006L,2007L,2008L,2009L,
                                           2010L,2011L,2012L,2013L,2014L,2015L,
                                           2016L,2017L,2018L,2019L,2020L,
                                           2021L,2022L,2023L,2024L,2025L),
```

```
        No..Reported.Pertussis.Cases = c(107473,164191,165418,
                                         152003,202210,181411,161799,197371,
                                         166914,172559,215343,179135,265269,
                                         180518,147237,214652,227319,
                                         103188,183866,222202,191383,191890,
                                         109873,133792,109860,156517,74715,
                                         69479,120718,68687,45030,37129,
                                         60886,62786,31732,28295,32148,
                                         40005,14809,11468,17749,17135,
                                         13005,6799,7717,9718,4810,3285,
                                         4249,3036,3287,1759,2402,1738,1010,
                                         2177,2063,1623,1730,1248,1895,
                                         2463,2276,3589,4195,2823,3450,
                                         4157,4570,2719,4083,6586,4617,
                                         5137,7796,6564,7405,7298,7867,
                                         7580,9771,11647,25827,25616,15632,
                                         10454,13278,16858,27550,18719,
                                         48277,28639,32971,20762,17972,
                                         18975,15609,18617,6124,2116,3044,
                                         7063,22538,21996)
       )
```

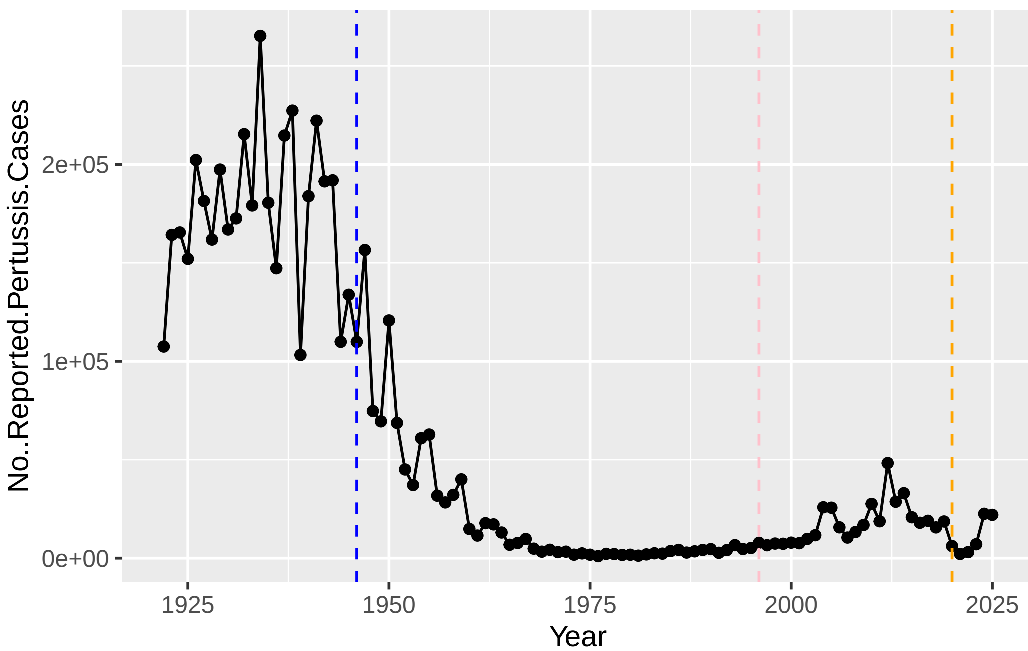
Q1: make a plot of year vs cases:

```
library(ggplot2)
ggplot(cdc,aes(Year, No..Reported.Pertussis.Cases))+
  geom_point()+
  geom_line()
```



Q. add some major milestones including the first wP vaccine roll-out (1946), the switch to the newer aP vaccine (1996), the COVID years (2020).

```r
library(ggplot2)
ggplot(cdc,aes(Year, No..Reported.Pertussis.Cases))+
  geom_point()+
  geom_line()+
  geom_vline(xintercept=1946, col="blue", lty=2)+
  geom_vline(xintercept=1996, col="pink", lty=2)+
  geom_vline(xintercept = 2020, col="orange", lty=2)
```



there were high case numbers in the pre-1940s.then the number of cases decreases until around 1996, where it then started rising in cases. change from wP vaccine to aP vaccine. and it is also aP vaccine in adolescence

key question to address: aP induced protection wanes faster than wP - why?

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

CMI PB project

computational models of immunity (https://www.cmi-pb.org/) - pertubisis boost project aims to provide the scientific community with this very information

They make there data avaliable via JSON format APIs. We can read this in R with the `read_json()` function from the **jsonlite** package:

```
library(jsonlite)

subject <- read_json("http://cmi-pb.org/api/v5_1/subject",
                     simplifyVector = TRUE)
head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q. How many wP and aP indiviudals are in this `subject` table?

```r
table (subject$infancy_vac)
```

```
aP wP
87 85
```

Q. what is the biological sex breakdown?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. in terms of race and gender is this dataset representative of the US population?

```r
table(subject$race, subject$biological_sex)
```

```
                                           Female Male
  American Indian/Alaska Native                 0    1
  Asian                                        32   12
  Black or African American                     2    3
  More Than One Race                           15    4
  Native Hawaiian or Other Pacific Islander     1    1
  Unknown or Not Reported                      14    7
  White                                        48   32
```

Let's read some more dataset tables

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen",
                      simplifyVector = TRUE)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer",
                      simplifyVector = TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

to analyse this data, we need to first “join”/merge/link the different tables so we all have the data in one place, not spread across different tables.

we can use the `*_join()` family of functions from **dplyr** to do this

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
```

```
3           1     IgG                TRUE      PT    68.56614       3.736992
4           1     IgG                TRUE     PRN   332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q. What Antibody isotypes are measured for these patients?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. what antigens are reported?

```r
table(abdata$antigen)
```

```

    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

Let's focus on the IgG isotype and make a plot of MFI_normalized for all antigens

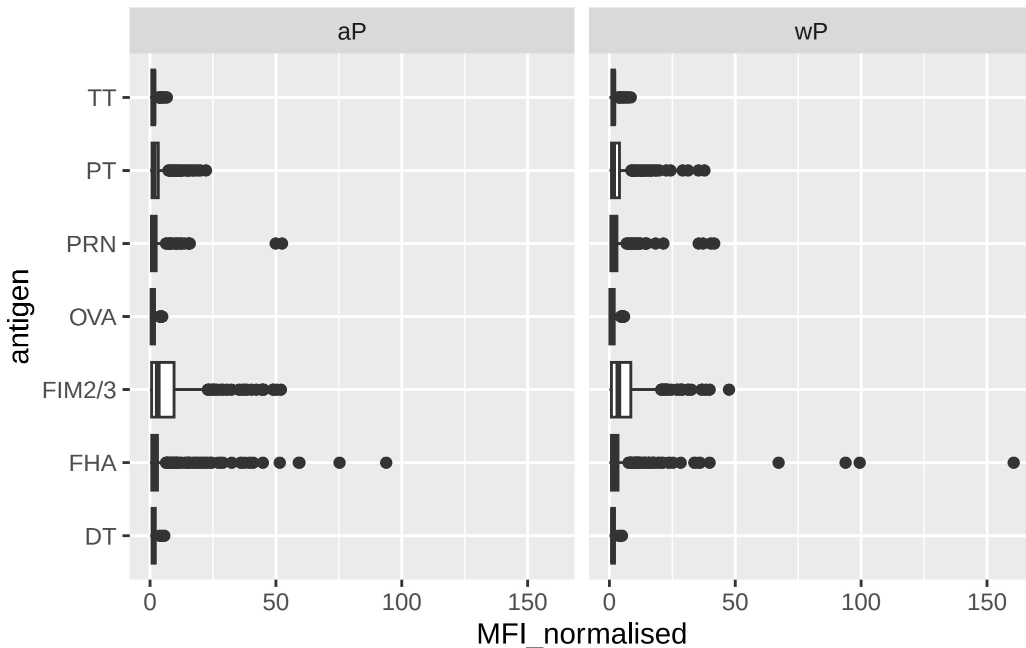
```r
igg <- abdata |>
  filter (isotype == "IgG")

head (igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IUI/ML                0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
```

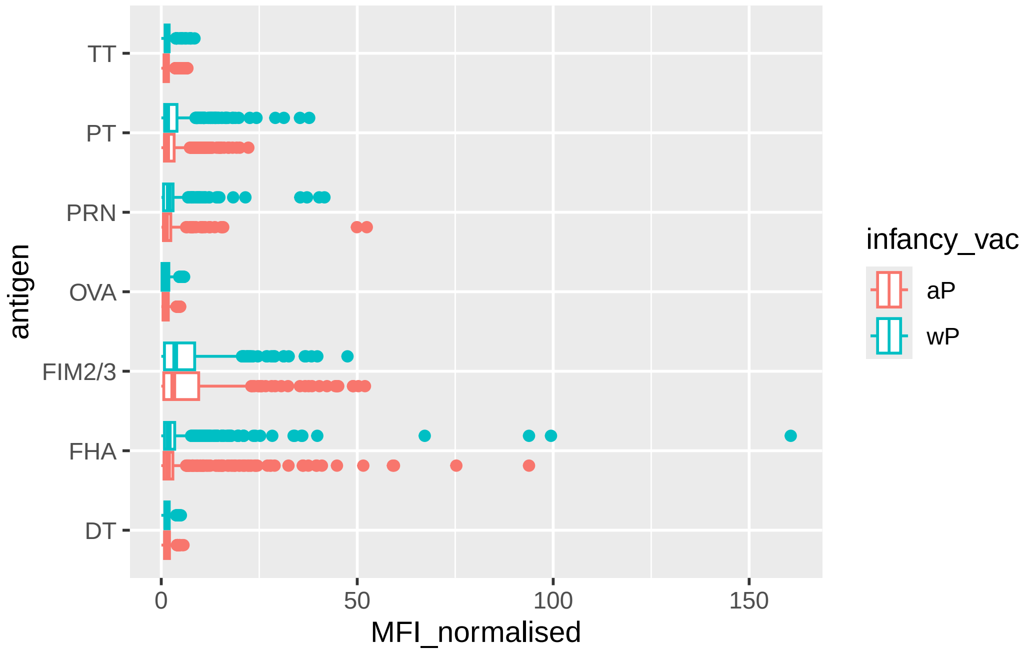
```
5                                      -3                                     0           Blood
6                                      -3                                     0           Blood
  visit
1       1
2       1
3       1
4       1
5       1
6       1
```

```
ggplot(igg)+
  aes(MFI_normalised, antigen)+
  geom_boxplot()+
  facet_wrap(~infancy_vac)
```



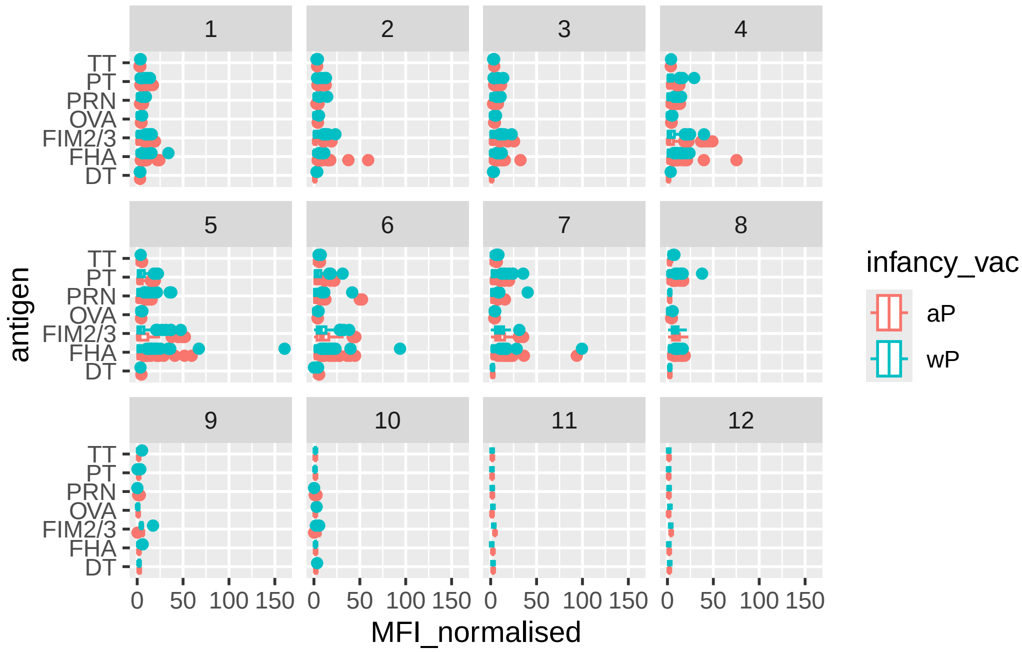
Q. is there a difference for aP vs wP individuals with these values?

```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()
```



Q. is there a temprol response - i.e do values increase or decrease over time?

```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~visit)
```

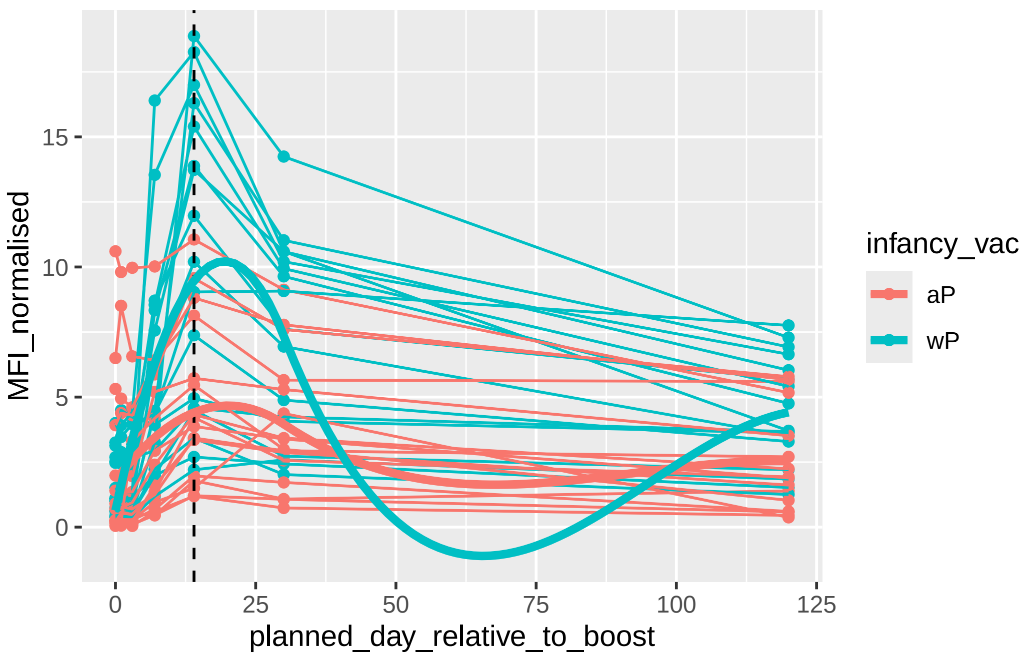


focus on "PT" pertusisis Toxin antigen

```
pt.igg.21 <- igg |> filter (antigen == "PT",
                dataset == "2021_dataset")
```

```
ggplot(pt.igg.21)+
  aes(planned_day_relative_to_boost, MFI_normalised,
      col = infancy_vac,
      group = subject_id)+
  geom_point()+
  geom_line()+
  geom_vline(xintercept = 14, col ="black", lty = 2)+
  geom_smooth(aes(group=infancy_vac),
              method = "loess",
              span = 1,
              se = FALSE,
              linewidth = 1.5
)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



MFI_normalised
15
10
5
0
0
25
50
75
100
125
planned_day_relative_to_boost
infancy_vac
aP
wP